Vision

https://excalidraw.com/#room=df22cb35c8ea1e701586,2Tz5deo5leUqiEQ1TUf67A

Requirements

https://wiki.mypaytm.com/display/RFC/Rule+Templatization+PRD

Design Details

https://link.excalidraw.com/l/7sT5uNtCIw3/3EYiKQzZEaY

Templatization Concepts

Modus Operandi

The specific methods or patterns used by fraudsters to carry out their activities. Understanding common modus operandi helps in creating more effective prevention strategies.

Template

A predefined set of templated rules, configurations, and workflows that can be applied to detect and prevent specific types of fraud or reserved event flows i.e. UPI transactions, PG transactions etc.. Templates can be customized and reused across different scenarios or business units.

Limits and Thresholds

Predetermined values or boundaries set within the system to trigger alerts or actions when exceeded. For example:

- Transaction amount limits
- Number of transactions per day

- Login attempts before account lockout

Environment

The specific context or setting in which the fraud prevention system operates. This could include:

- Development environment for testing new rules
- Staging environment for final validation
- Production environment for live operations

Instance Tenants

Separate, isolated instances of the fraud prevention system for different clients or business units, allowing for customized configurations while maintaining a shared core system.

Catalog

A comprehensive collection of fraud prevention rules, algorithms, and models that can be applied to different scenarios. The catalog serves as a repository of fraud detection methods or reserved event flows

Decision Flow

The sequence of steps and logic applied to evaluate a transaction or action for potential fraud. This typically involves multiple decision points and can branch based on various criteria.

Decision Node

A specific point in the decision flow where a particular check or evaluation is performed. Each node can lead to different outcomes based on the result of its evaluation.

Lists

Curated collections of entities (e.g., users, IP addresses, device IDs) used for quick reference in fraud detection:

- Whitelist: Trusted entities that are pre-approved or require less scrutiny
- Blacklist: Known fraudulent or high-risk entities that are blocked or require additional verification

Global and Local Configs

- Global Configs: System-wide settings that apply across all tenants or instances
- Local Configs: Instance-specific settings that override or complement global configs for customized fraud prevention

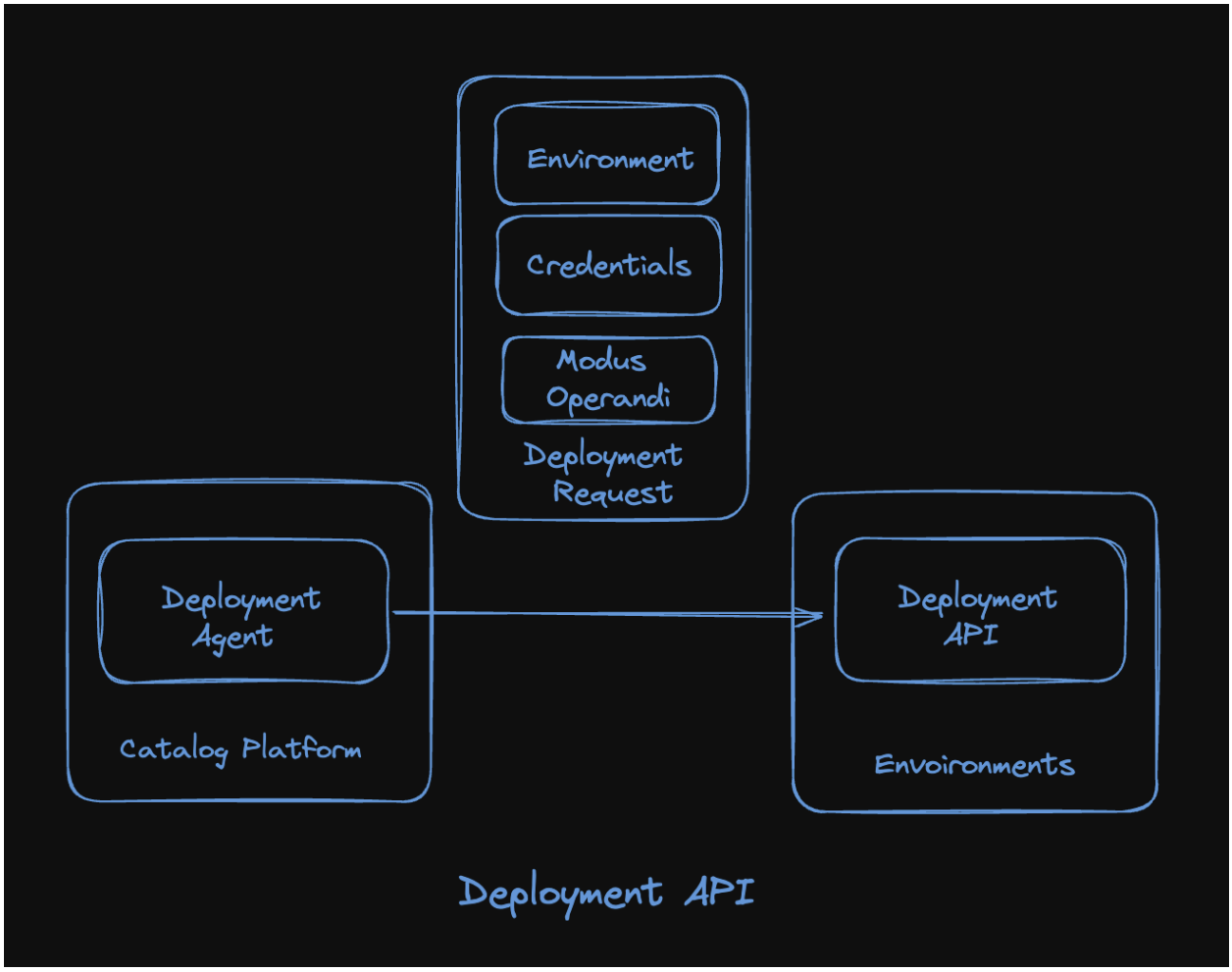
Roles and Permissions

A system of access control that defines what actions different users can perform within the fraud prevention system:

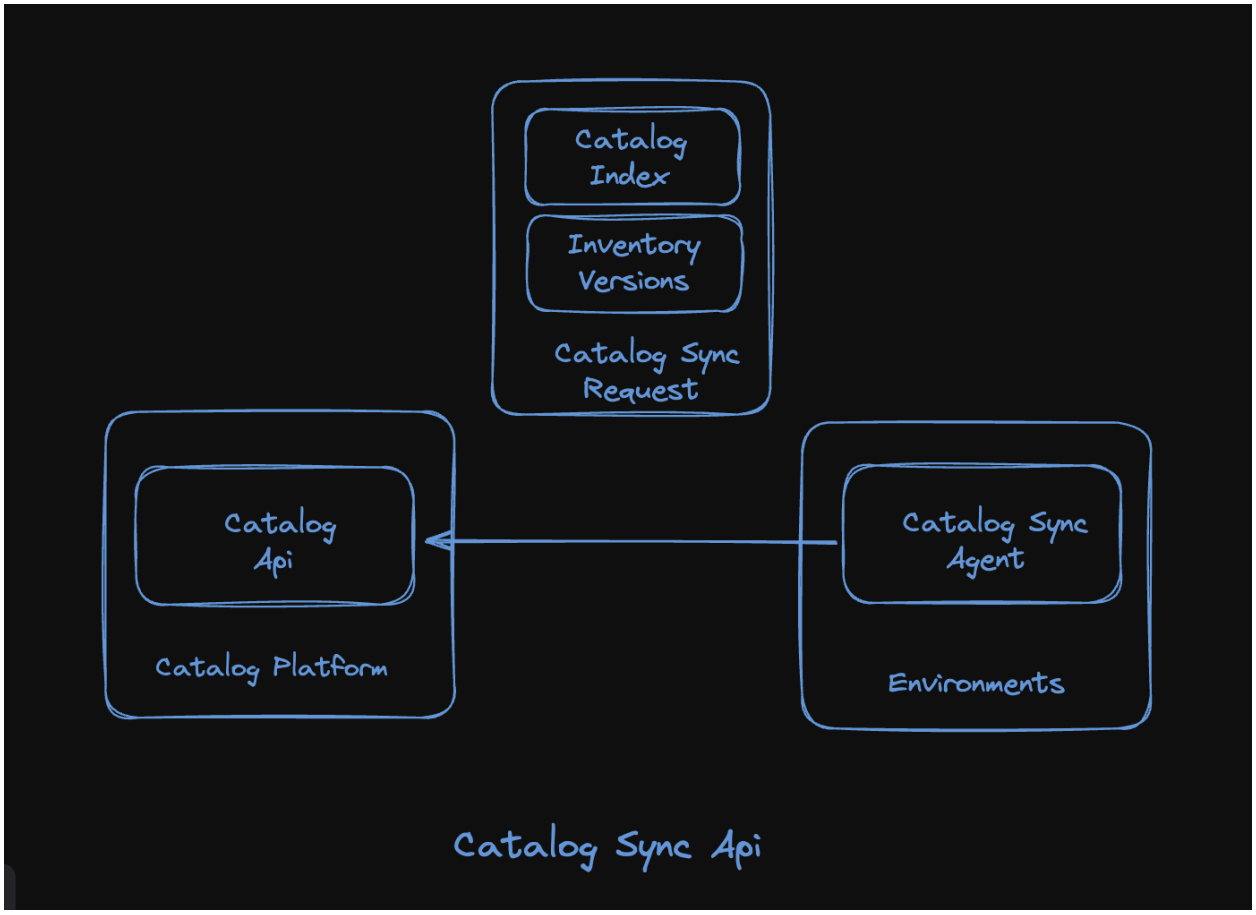
- Admin: Full system access and configuration rights
- Analyst: Ability to review and investigate flagged transactions
- Read-only: Access to view data but not make changes
- Custom roles with specific permissions (which will be defined later owing to client/business needs)

High Level Api Surfaces

Template Deployment

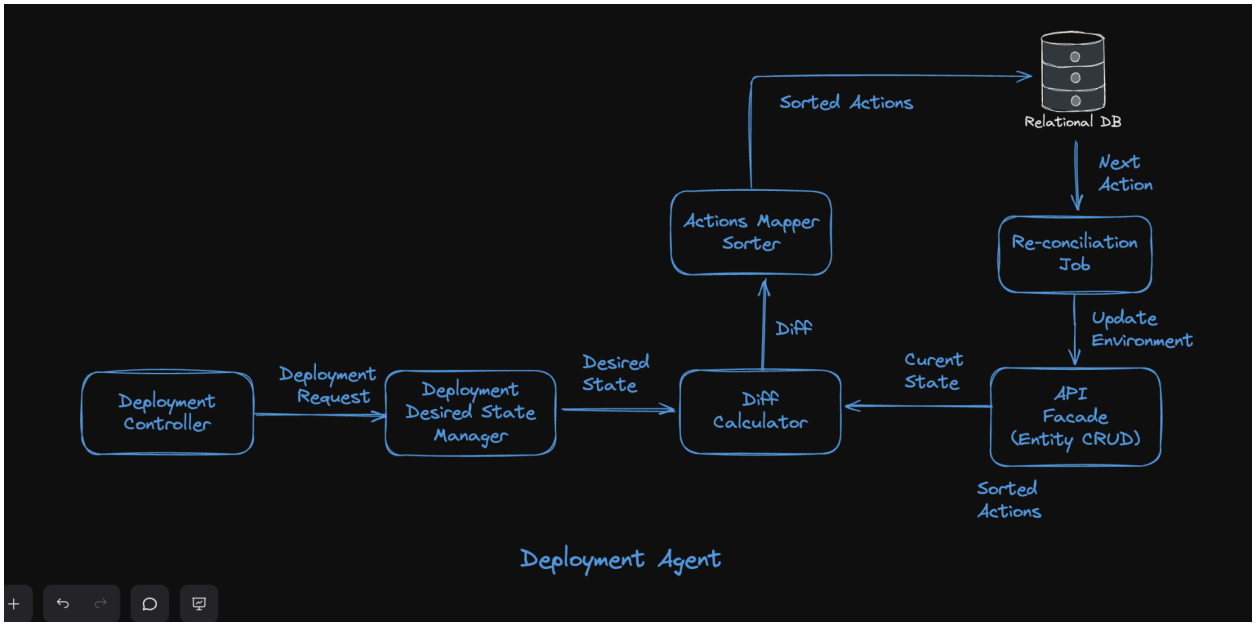


Catalog Sync

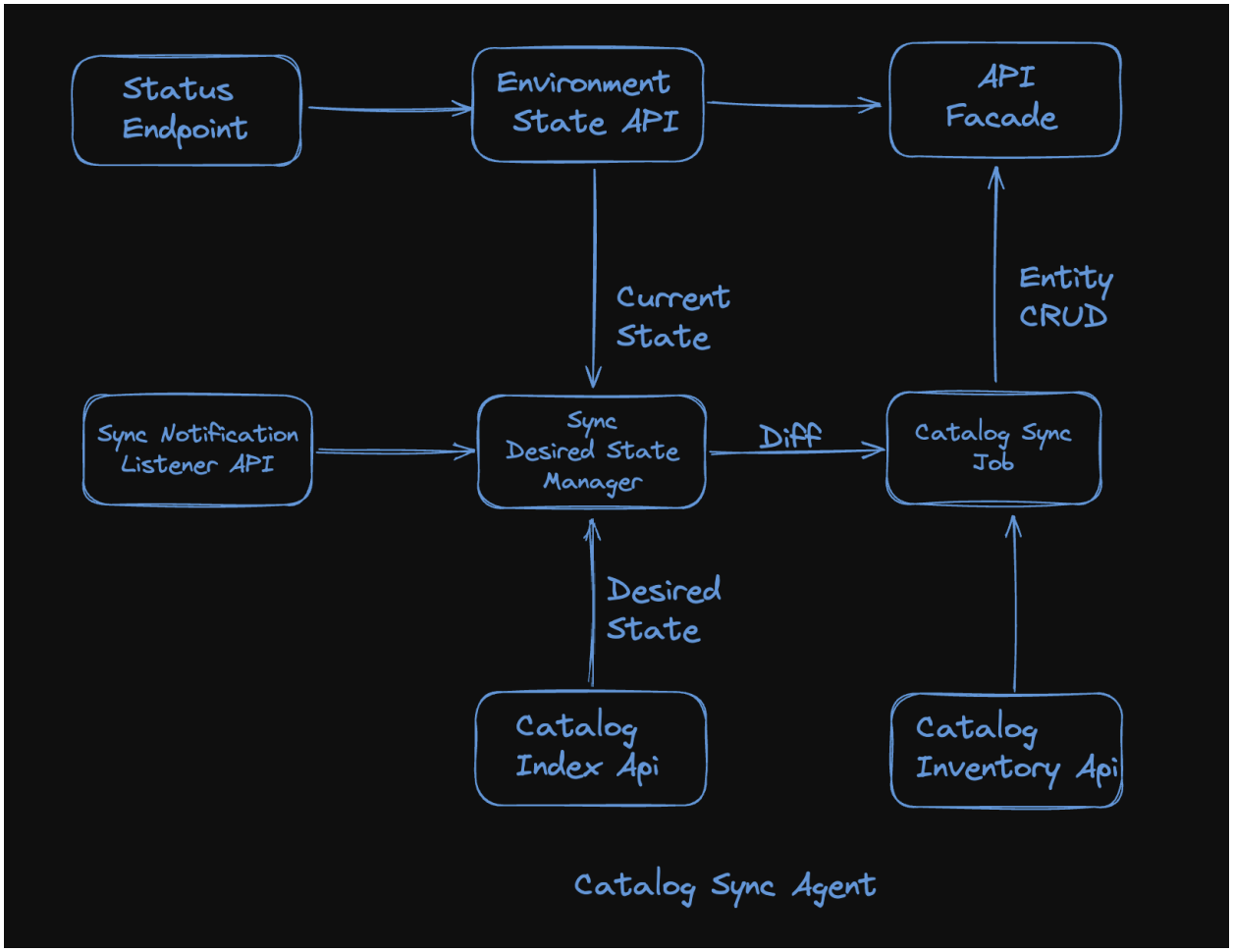


Component Level Interaction

Deployment Workflow



Catalog Sync Workflow



Detailed Design

DSM Framework

TBD

Diff Calculator

TBD

Action Sorter and Mapper

TBD

Template to Entity CRUD Mapper

TBD